

Analytical Note: Parity Conservation and Cavity Non-Thermalization in the Periodically Driven LMG–Cavity System

Analytic discussion for DMBL-Engine project

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1 The Hamiltonian

The system under study is a periodically driven Lipkin–Meshkov–Glick (LMG) model coupled to a single bosonic cavity mode:

$$H(t) = -\frac{4J}{N} S_z^2 \otimes I_{\text{ph}} + 2h(t) S_x \otimes I_{\text{ph}} + \hbar\omega_0 I_{\text{spin}} \otimes a^\dagger a + \frac{2g}{\sqrt{N}} (a + a^\dagger) \otimes S_x, \quad (1)$$

where $h(t) = h_0 \cos(\omega t)$, $S_\alpha = \sum_i \sigma_\alpha^{(i)}/2$ are collective spin- S operators with $S = N/2$, and $a, a^\dagger$ are photon ladder operators. The system is initialized in $|\psi_0\rangle = |S_x=+S\rangle \otimes |0\rangle_{\text{ph}}$.

2 Exact Z_2 Symmetry of the Full Driven Hamiltonian

2.1 The conserved parity

Define the spin-rotation operator

$$\Pi_x = e^{i\pi S_x} \otimes I_{\text{ph}}. \quad (2)$$

This is a rotation of the spin by π about the x -axis, leaving the photon sector untouched. Under Π_x :

$$S_x \longrightarrow S_x, \quad S_y \longrightarrow -S_y, \quad S_z \longrightarrow -S_z. \quad (3)$$

Therefore $S_z^2 \rightarrow S_z^2$, $S_x \rightarrow S_x$, and $(a + a^\dagger) \rightarrow (a + a^\dagger)$. Checking each term of $H(t)$:

$$\Pi_x \left(-\frac{4J}{N} S_z^2 \right) \Pi_x^\dagger = -\frac{4J}{N} (-S_z)^2 = -\frac{4J}{N} S_z^2 \quad \checkmark \quad (4)$$

$$\Pi_x (2h(t) S_x) \Pi_x^\dagger = 2h(t) S_x \quad \checkmark \quad (5)$$

$$\Pi_x (\hbar\omega_0 a^\dagger a) \Pi_x^\dagger = \hbar\omega_0 a^\dagger a \quad \checkmark \quad (6)$$

$$\Pi_x \left(\frac{2g}{\sqrt{N}} (a + a^\dagger) S_x \right) \Pi_x^\dagger = \frac{2g}{\sqrt{N}} (a + a^\dagger) S_x \quad \checkmark \quad (7)$$

Hence

$$\boxed{[\Pi_x, H(t)] = 0 \quad \text{for all } t.} \quad (8)$$

2.2 Consequence for the initial state

The eigenvalues of $\Pi_x = e^{i\pi S_x}$ on S_x eigenstates $|m_x\rangle$ are $e^{i\pi m_x} = (-1)^{m_x}$ (for integer m_x with S integer). The initial spin state $|S_x=+S\rangle$ has eigenvalue $(-1)^S = (-1)^{N/2}$. For $N = 400$ this is $(-1)^{200} = +1$.

Because $[\Pi_x, H(t)] = 0$ exactly, the parity eigenvalue is a constant of motion. The system is permanently confined to the *even- m_x* sector: $m_x \in \{0, \pm 2, \pm 4, \dots, \pm S\}$ (for $N/2$ even). The $\binom{N+1}{2}$ odd- m_x spin states are *permanently dark*.

3 The Dicke Parity Is Broken by the Drive

The standard Dicke-model parity operator is

$$\Pi_D = e^{i\pi(a^\dagger a + J + J_z)}. \quad (9)$$

Under Π_D , one finds $J_\pm \rightarrow -J_\pm$ (since J_z shifts by ± 1), so $S_x = (J_+ + J_-)/2 \rightarrow -S_x$, while $e^{i\pi a^\dagger a}(a + a^\dagger)e^{-i\pi a^\dagger a} = -(a + a^\dagger)$. Thus the coupling term $(a + a^\dagger)S_x \rightarrow -(a + a^\dagger)(-S_x) = (a + a^\dagger)S_x$ is invariant. However, the drive transforms as

$$2h(t)S_x \xrightarrow{\Pi_D} -2h(t)S_x, \quad (10)$$

so $[\Pi_D, H(t)] \neq 0$ whenever $h(t) \neq 0$. *The Dicke parity is broken by the transverse drive.*

4 The Coupling Is a QND Interaction for S_x

The coupling Hamiltonian commutes with the spin observable it couples to:

$$[H_{\text{coup}}, I_{\text{ph}} \otimes S_x] = \frac{2g}{\sqrt{N}}(a + a^\dagger) \otimes [S_x, S_x] = 0. \quad (11)$$

This is a *quantum-non-demolition* (QND) property: the coupling does *not* change the S_x quantum number of the spin. In the Heisenberg picture:

$$\dot{S}_x \Big|_{H_{\text{coup}}} = i[H_{\text{coup}}, S_x] = 0. \quad (12)$$

The spin's S_x dynamics is driven *solely* by $H_{\text{LMG}}(t)$. The cavity has *no back-action* on S_x .

4.1 Consequence: the cavity equation of motion

In the Heisenberg picture the photon field satisfies

$$\dot{a} = -i\omega_0 a - i\frac{2g}{\sqrt{N}} S_x(t), \quad (13)$$

where $S_x(t)$ evolves *independently* of a (due to the QND property). Equation (??) is that of a *harmonically oscillator driven by an external force* $F(t) \propto S_x(t)$. The cavity mode can be displaced and can accumulate coherent amplitude, but the driving force is not truly dissipative in the quantum-thermodynamic sense.

4.2 Why coherent driving does not thermalize the cavity

For the cavity to thermalize to temperature β^{-1} , the coupling must provide both *heating* (stimulated absorption) and *cooling* (stimulated emission) with rates satisfying detailed balance. These rates are set by the bath spectral function $\mathcal{S}(\nu) = \int_{-\infty}^{\infty} dt e^{i\nu t} \langle S_x(t) S_x(0) \rangle_{\text{bath}}$.

Because the coupling conserves S_x and the driving is Hamiltonian (not dissipative), the cavity equation of motion (??) admits only *unitary* dynamics. An initially pure state remains pure. More concretely, starting from the vacuum $|0\rangle_{\text{ph}}$, the photon evolves into a *coherent state* that tracks the spin:

$$\alpha(t) = \langle a \rangle(t) = -i \frac{2g}{\sqrt{N}} \int_0^t dt' \langle S_x(t') \rangle e^{-i\omega_0(t-t')}. \quad (14)$$

Even if $\langle S_x \rangle_{\text{time-avg}} = 0$ (infinite-temperature spin state), the photon state fluctuates around the vacuum but does not build up a thermal occupation. True thermalization requires the bath to provide *quantum noise* with the right spectral properties; for the QND coupling this requires the many-body fluctuations of S_x to act as a proper Markovian heat bath. For the finite LMG system this condition is generically *not* met: the correlation function $\langle S_x(t) S_x(0) \rangle$ is dominated by a few Floquet frequencies (it does not decay to zero), so the spectral function $\mathcal{S}(\nu)$ is a comb of delta functions rather than a smooth function of ν .

5 Summary and Physical Interpretation

1. **Exact conserved parity (but acts only on spins):** $\Pi_x = e^{i\pi S_x} \otimes I_{\text{ph}}$ is conserved for all t . The initial state has parity +1; only even- m_x spin states are ever populated. This halves the effective spin Hilbert space but does not directly restrict which photon states can be reached.
2. **Broken Dicke parity:** The photon-spin Dicke parity $\Pi_D = e^{i\pi(a^\dagger a + J + J_z)}$ is broken by the drive $2h(t)S_x$. At low drive frequency the Floquet Hamiltonian $\approx$ time-averaged H *does* approximately conserve Π_D , which forces the system to stay near the initial Dicke parity sector during most of the slow cycle.
3. **QND coupling is the primary obstruction:** $[H_{\text{coup}}, S_x] = 0$ means the cavity drives the spin's S_z and S_y components but *never changes* S_x . The coupling is a displacement of the photon conditioned on S_x , not a genuine energy-exchange channel.
4. **Remedy:** Replace or augment the coupling with one that does not commute with the relevant spin observable, e.g., $(a^\dagger J_- + a J_+)$ (Jaynes–Cummings type) or $(a + a^\dagger)S_z$. The former exchanges quanta resonantly and satisfies detailed balance; it is the standard mechanism for cavity thermalization. With the present $(a + a^\dagger)S_x$ coupling, the Jaynes–Cummings energy-exchange channel is absent in the rotating-wave approximation.

6 Conclusion

Parity conservation *is* relevant: the Π_x symmetry is exact and confines the spin to the even- S_x sector, while the Dicke parity Π_D is only approximately conserved at low drive

frequency. However, the more fundamental reason the cavity fails to thermalize is that the coupling $H_{\text{coup}} \propto (a + a^\dagger)S_x$ is a QND interaction: it conserves S_x and provides only coherent displacements of the photon. Without a genuine energy-exchange channel (or a truly Markovian spin bath in the thermodynamic limit), the cavity cannot be driven to infinite effective temperature by the spin alone.